but not in their chemical properties.

Bohr and Wheeler, building on the work of refugee physicists Lise Meitner and her nephew Otto Frisch, found that the two isotopes of uranium have very different nuclear properties. Specifically, hitting a U-235 nucleus with a neutron leads the nucleus to *fission*: it splits into two smaller nuclei, releasing a fabulous quantity of energy, along with a few free-floating neutrons. With enough U-235—a critical mass—the neutrons left over from fission will hit more U-235 nuclei, which will split in turn, releasing even more neutrons and starting a chain reaction. Left uncontrolled in 120 pounds of pure U-235—a small sphere of the dense metal, less than twenty centimeters across—a nuclear chain reaction would explode with the power of 15,000 tons of TNT, enough to instantly level a small city. Controlling the reaction by absorbing some of the excess neutrons would allow you to power a small city instead, for days on end, with the same 120 pounds of U-235.

U-238 is a different story. Those three extra neutrons give it a little more stability, so hitting it with a neutron won't split it as easily. This makes it impossible to build a bomb out of U-238. And fortunately, about 99.3 percent of uranium in nature is U-238. To build an atomic bomb, you would need to separate the tiny quantity of U-235 from the enormous bulk of U-238—and since they're chemically identical, the only way to separate them is to take advantage of the fact that U-238 is 1.3 percent heavier than U-235. This guaranteed that nuclear power would be phenomenally difficult to achieve, requiring enormous quantities of uranium and city-sized industrial diffusion and centrifuge facilities. "It can never be done unless you turn the United States into one huge factory," concluded Bohr.

Yet the risk of not pursuing nuclear power was too high. If Nazi Germany were to build an atomic bomb, the war would be over. The Einsteins and Fermis and Borns of the world would never be able to escape Hitler's Reich. "A little bomb like that," said Fermi, cupping his hands as he looked out over Manhattan, "and it would all disappear."

"Can you guess where I found out about [fission]? In… the infirmary."